Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**LAB – Reinforcement Learning**
Semester	6th	**Course Code**	**23ADS1644**
Roll No	**42**	**Name of Student**	**Krishna Mishra**

Practical Number	**06**
Aim	To implement the Q-Learning algorithm to enable an agent to learn optimal actions through trial and error.
Problem Definition	To implement the Q-Learning algorithm that enables an agent to learn optimal actions in an environment through trial and error, without prior knowledge of the environment.
Theory (100 words)	Q-Learning is a model-free reinforcement learning algorithm that helps an agent learn the best actions to take in a given environment by interacting with it. The agent learns a policy that maximizes cumulative rewards over time using a Q-table, which stores values representing the expected reward for taking a specific action in a given state. The core idea of Q-Learning is based on the Bellman equation, where Q-values are updated iteratively using the formula: $Q(s, a) = Q(s, a) + \alpha [r + \gamma \max Q(s', a') - Q(s, a)]$ Here, α is the learning rate, γ is the discount factor, and r is the reward. The agent follows an exploration-exploitation strategy, often using epsilon-greedy, where it sometimes explores new actions and sometimes exploits known optimal actions. Over time, the Q-values converge to optimal values, allowing the agent to make better decisions. Q-Learning is widely used in robotics, gaming, and decision-making systems due to its simplicity and effectiveness in unknown environments.

Procedure and Execution (100 Words)	Algorithm: 1. Define the environment with states, actions, and rewards. 2. Initialize the Q-table with zeros. 3. Set parameters: learning rate (α), discount factor (γ), and exploration rate (ε). 4. For each episode, initialize the starting state. 5. Choose an action using epsilon-greedy strategy. 6. Perform the action and observe reward and next state. 7. Update Q-value using the Q-Learning formula. 8. Repeat until the goal state is reached or episode ends. 9. Evaluate learned policy.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

n_states = 16
n_actions = 4
goal_state = 15

Q_table = np.zeros((n_states, n_actions))

learning_rate = 0.8
discount_factor = 0.95
exploration_prob = 0.2
epochs = 1000

def get_next_state(state, action):
  row, col = divmod(state, 4)

  if action == 0 and col > 0:
```

```
    col -= 1
  elif action == 1 and col < 3:
    col += 1
  elif action == 2 and row > 0:
    row -= 1
  elif action == 3 and row < 3:
    row += 1

  return row * 4 + col

q_values_grid = np.max(Q_table,
axis=1).reshape((4, 4))

plt.figure(figsize=(6, 6))
plt.imshow(q_values_grid, cmap='coolwarm',
interpolation='nearest')
plt.colorbar(label='Q-value')
plt.title('Learned Q-values for Each State')
plt.xticks(np.arange(4), ['0', '1', '2', '3'])
plt.yticks(np.arange(4), ['0', '1', '2', '3'])
plt.gca().invert_yaxis()
plt.grid(True)

for i in range(4):
    for j in range(4):
        plt.text(j, i, f'{q_values_grid[i,
j]:.2f}', ha='center', va='center',
color='black')

plt.show()

print("Learned Q-table:")
print(Q_table)

import random
```

```
def choose_action(state, Q_table,
exploration_prob):
  if random.uniform(0, 1) < exploration_prob:
    return random.randint(0, n_actions - 1) #
Explore: choose a random action
  else:
    return np.argmax(Q_table[state, :]) #
Exploit: choose the action with the highest Q-
value

def get_reward(state, goal_state):
  return 1 if state == goal_state else 0

# Q-learning training loop
for episode in range(epochs):
  current_state = 0 # Start at state 0 for
each episode
  done = False

  while not done:
    action = choose_action(current_state,
Q_table, exploration_prob)
    next_state = get_next_state(current_state,
action)
    reward = get_reward(next_state,
goal_state)

    # Q-value update formula
    old_q_value = Q_table[current_state,
action]
    next_max_q_value =
np.max(Q_table[next_state, :])
    new_q_value = old_q_value + learning_rate
* (reward + discount_factor * next_max_q_value
- old_q_value)
```

```
    Q_table[current_state, action] =
new_q_value

    current_state = next_state

    if current_state == goal_state: # or
episode reaches a certain length
      done = True

print("Q-learning training complete.")
```

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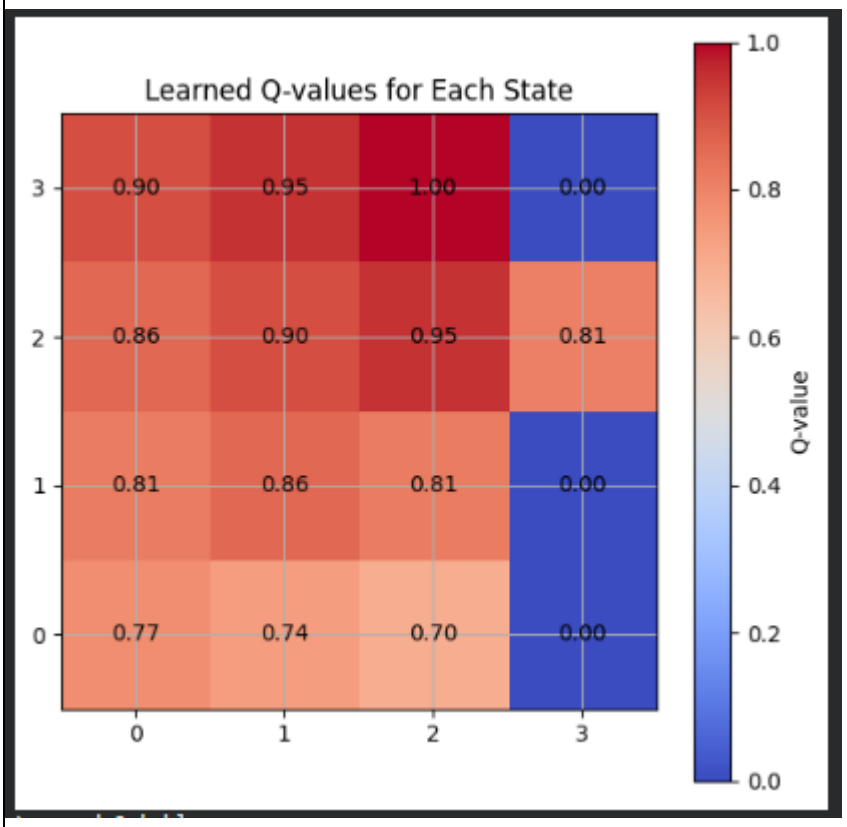
Output:




```
Learned Q-table:
[[0.73509189 0.6983373  0.73509189 0.77378094]
 [0.73509189 0.64871233 0.69275031 0.651605  ]
 [0.6981124  0.         0.         0.        ]
 [0.         0.         0.         0.        ]
 [0.77378094 0.81450625 0.73509189 0.81450625]
 [0.77378094 0.77257679 0.69833722 0.857375  ]
 [0.81447916 0.         0.52646328 0.651605  ]
 [0.         0.         0.         0.        ]
 [0.81450625 0.857375   0.77378094 0.857375  ]
 [0.81450625 0.90249999 0.81450625 0.9025    ]
 [0.857375   0.67399414 0.77294335 0.95      ]
 [0.80544894 0.44693586 0.         0.        ]
 [0.857375   0.9025     0.81450625 0.857375  ]
 [0.857375   0.95       0.857375   0.9025    ]
 [0.9025     1.         0.90249999 0.95      ]
 [0.         0.         0.         0.        ]]
```

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Conclusion	Q-Learning enables an agent to learn optimal actions through continuous interaction with the environment. It effectively improves decision-making by maximizing long-term rewards without requiring a model of the environment.
Date	**28-03-2026**

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